F18CF – Linear Algebra

Tutorial Sheet 1: Systems of linear equations

Getting started.

1. For each of the following systems of linear equations, determine the number of pivot and free variables, and find all the solutions, and double-check your answers:

$$\text{(a)} \begin{cases} x+2y-z &= 2 \\ 2x+y+z &= 3 \\ x-y+2z &= 2 \end{cases}, \quad \text{(b)} \begin{cases} x+y+z &= 1 \\ x+2y+3z &= 3 \\ x+4y+9z &= 7 \end{cases}, \quad \text{(c)} \begin{cases} x-2y+z &= 1 \\ 2x-y-3z &= 2 \\ 3x-7z &= 3 \end{cases}.$$

2. For each of the following systems of linear equations, determine the number of pivot and free variables, find all the solutions, and double-check your answers:

$$\text{(a)} \begin{cases} x+y-z+u &= 0 \\ 2x+y+2z-u &= 0 \\ 4x+3y+u &= 0 \\ 5x+3y+3z-u &= 0 \end{cases}, \quad \text{(b)} \begin{cases} 3x+y+z+u &= 12 \\ x+3y+z+u &= 0 \\ x+y+3z+u &= 12 \\ x+y+z+3u &= 0 \end{cases}$$

Problems.

3. **An optimisation problem.** A company would like to start a bike rental business in Edinburgh, with parking spots scattered in four main zones: Old Town (zone 1), Haymarket (zone 2), Fountainbridge (zone 3), and Newington (zone 4). During a trial period, the company installed 200 bikes across the city and monitored their position to understand how customers use them. They noticed the following:

   - On average, 50% of bikes in zone 1 are parked back in zone 1 at the end of the day, 20% are parked in zone 2, 20% in zone 3, and 10% in zone 4.
   - On average, 20% of bikes in zone 2 are parked back at zone 2 at the end of the day, 40% are parked in zone 1, 20% in zone 3, and 20% in zone 4.
   - On average, 40% of bikes in zone 3 are parked back in zone 3 at the end of the day, 30% are parked in zone 1, 20% in zone 2, and 10% in zone 4.
   - On average, 50% of bikes in zone 4 are parked back in zone 4 at the end of the day, 10% are parked in zone 1, 20% in zone 2, and 20% in zone 3.

   (a) Find the *stable distribution* of bikes, that is, the distribution of bikes so that the number of bikes parked in each zone at the beginning of a day remains unchanged at the end of that day.

We are interested in a more general situation. We denote by $a_{ij} \in [0,1]$ the proportion of bikes in zone j being parked in zone i at the end of the day. Consider again the homogeneous system of four linear equations in four variables encoding the fact that, at the end of each day, the number of bikes is the same as at the beginning of that day. Consider its associated matrix, and denote by $R_1, \ldots, R_4$ its rows.

(b) Show that $R_1 + R_2 + R_3 + R_4$ is a row consisting solely of zeroes. Deduce that the system has infinitely many solutions.

4. **Polynomial interpolation.** Determine whether the following polynomial functions exist:

(a) A polynomial $P(x) = ax^2 + bx + c$ such that $P(1) = 3$, $P(2) = 3$, $P(3) = 5$, and $P(-1) = 6$.

(b) A polynomial $P(x) = ax^2 + bx + c$ such that $P(-1) = 1$, $P(1) = -1$, $P'(1) = 3$, and $P(2) = 4$.

5. **The geometry of solutions of systems of linear equations.**

(a) Consider a linear equation of the form

$$a_1x_1 + \cdots + a_nx_n = b,$$

with $a_1, \ldots, a_n, b \in \mathbb{R}$. Show that if $(u_1, \ldots, u_n)^T$ and $(v_1, \ldots, v_n)^T$ are two distinct solutions of this equation, then for every $\lambda \in \mathbb{R}$,

$$(\lambda u_1 + (1-\lambda)v_1, \ldots, \lambda u_n + (1-\lambda)v_n)^T$$

is also a solution. What does it mean geometrically for the solution set?

(b) Use (a) to recover the fact that a system of linear equations has either zero, one, or infinitely many solutions.

(c) Below are some subsets of $\mathbb{R}^3$.

i. a line in $\mathbb{R}^3$,

ii. the union of two arbitrary lines of $\mathbb{R}^3$, and

iii. a plane in $\mathbb{R}^3$.

For each of them, explain why they cannot be the set of solutions of a system of linear equations in three variables, or give an example of a system of linear equations in three variables whose set of solutions is the given subset.

6. **Equations with parameters.**

(a) Find a condition on the parameters $a, b, c \in \mathbb{R}$ so that the following system of linear equations is inconsistent:

$$\begin{cases} 2x_1 - x_2 + 3x_3 &= a \\ 3x_1 + x_2 - 5x_3 &= b \\ -5x_1 - 5x_2 + 21x_3 &= c \end{cases}$$

(b) Discuss the number of solutions of the following system of linear equations depending on the value of the parameter $a \in \mathbb{R}$:

$$\begin{cases} x_1 + x_2 - x_3 & = & 2 \\ x_1 + 2x_2 + x_3 & = & 3 \\ x_1 + x_2 + (a^2 - 5)x_3 & = & a \end{cases}$$